



AI Red Team vs. Blue Team Bootcamp — Build Challenge

3-Hour Hands-On Attack Demo Session · Jun 11th, 2026

Overview

This session splits participants into six teams over a three-hour session: three Attacker (Red) teams and three Defender (Blue) teams. Each team has 2 hours to build a tool. The Red teams build a generator that produces realistic process-command attacks on demand; the Blue teams build a detection system that hunts them down. At the demo, the judges name a scenario, a Red team's tool generates the attack live, and that output becomes the input to a Blue team's system. Scoring determines the winner of each match.

The entire challenge is scoped to process-level activity only. No network, no infrastructure, no credentials — just process commands.

The Setup

	Red Team (Attackers)	Blue Team (Defenders)
Goal	Build a tool that generates a realistic process-command dataset hiding malicious activity, on demand for any scenario the judges name.	Build a tool that identifies which commands in the dataset are malicious and reconstructs the attack story.
Output	A working generator that, given a scenario, emits a CSV of ~220 process commands (20 malicious) plus a written attack story.	A list of flagged commands with a short story explaining the attack, produced as fast as possible.
Weapon	LLM-driven tool to generate, vary and blend commands and craft an attack narrative.	Regex, pattern matching, allowlists/blocklists, heuristics and AI to detect anomalies.
Win condition	Minimize defender true positives while maximizing evasion.	Maximize true positives, minimize false positives, and finish fast.



Red Team: Attacker Mission

What to Build

Build a tool that generates attacks. The tool — not a hand-written file — is the deliverable. At demo time the judges will name a scenario (for example: ransomware, crypto miner, lateral movement), and your tool must generate a fresh attack dataset for that scenario on the spot.

Input: your choice (a scenario prompt, flags, or a short description).

Output: a CSV file containing approximately 220 process commands. Each generated dataset must include:

- 20 malicious commands that simulate a real attack scenario (lateral movement, persistence, data exfiltration, privilege escalation, etc.).
- Approximately 200 benign commands as background noise.
- A short written attack story explaining what the generated attack does.

Attack Scenario

Your scenario must tell a coherent attack story — a beginning, middle, and end. For example: initial access via a scheduled task, then privilege escalation, then credential dumping, then exfiltration. The scenario does not need to be complex, but it must be logical and explainable. The tool should be able to produce a logical, consistent attack for whatever scenario the judges request — not just replay a single canned example.

At the demo, you will present your tool and the generated attack to the room. You should be able to explain what the attack was, why each command was chosen, and how you tried to hide it.

CSV Format

Column	Description
process_name	The name of the process (e.g. cmd.exe, python3, bash).
command_line	The full command line including arguments.
label	'malicious' or 'benign'. Stripped from the scored CSV; the values are captured in the shared ground-truth list.

Separately from the CSV, the Red Team submits a labeled list of its 20 malicious commands as the agreed ground truth for scoring. This list is shared openly with everyone, including the Blue Team. Because detection is automated and tool-based, seeing the list gives no real advantage, and it removes any ambiguity about which commands count as malicious.

Rules

- Exactly 20 malicious commands per generated dataset — no more, no less.



- Total dataset: approximately 220 rows (around 200 benign).
- Commands must be realistic: real process names, real argument syntax.
- Use the LLM to generate, vary and blend commands — not just to copy a list.
- The tool must regenerate a valid attack for any scenario the judges name, every time.
- Submit the separate labeled list of the 20 malicious commands to the judges at demo time.

Blue Team: Defender Mission

What to Build

Build a detection system tool that takes a CSV of process commands as input and outputs a verdict for each row: malicious or benign. You are free to choose your approach. Common building blocks include:

- Regular expressions and pattern matching on command-line arguments.
- Process-name allowlists and blocklists.
- Heuristic scoring (e.g. argument length, special-character density, known LOLBin names).
- An AI pass to classify commands or to verify and reduce false positives.

Before the session, the facilitator will provide a training dataset — a labeled sample of process commands you can use to calibrate your detection logic. This is separate from the attacker's final CSV.

The Story

At the demo, the Blue Team must present a short attack narrative based only on the commands they flagged as malicious. What happened? In what order? What was the attacker trying to achieve? The story is built entirely from your own detections — not from the attacker's explanation.

Bonus: Identify the Attack Scenario

If your reconstructed story correctly matches the attacker's actual scenario, your team earns +20 bonus points.

Scoring System

Scoring is based on the defender's performance against the attacker's labeled ground truth. Both teams are scored — defenders for detection quality and speed, attackers for evasion.

Event	Defender Score	Attacker Score
True Positive (malicious flagged correctly)	+1	-1



False Positive (benign flagged as malicious)	-1	+1
True Negative (benign correctly ignored)	0	0
False Negative (malicious missed)	0	+1
Defenders correctly reconstruct attack scenario	+20	0
Judges' score on tool UI / pitch	+10	+10
Detection speed (time to complete scan)	-1 for each second passed	0
AI-only detection (no pattern matching)	scored lower	—

Both teams start at 0. The team with the highest score at the end wins.

Demo & Judging Format

Each team gives a 5-minute pitch of its tool to the room. The judges name an attack scenario, the Red Team's tool generates the dataset live, and that output is fed into the Blue Team's detection system. The judges take a minute or two to deliberate and decide which tool is stronger, factoring in detection accuracy, speed, the quality of the attack story, and the tool itself — not just the head-to-head numbers.

Session Timeline

Phase	Duration	Red Team	Blue Team
Kickoff	15 min	Read brief, define attack-generator concept.	Read brief, plan detection architecture.
Build	115 min	Build and refine the generator tool using the LLM.	Build and test the detection system against training data.
Scoring and Demo	60 min	5-min pitch; generate the attack live for the judges' scenario; present the story.	5-min pitch; run detector against the generated CSV; present findings and story guess.



You craft the attack. They hunt for it. May the best team win.